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(57) Abstract :
Introduction: Oroxylin A is primarily sourced from the roots of Scutellaria baicalensis, a medicinal plant commonly used in traditional Chinese medicine. It can also be found in other Scutellaria species. The plant's rich bioactive profile makes it a significant source of various flavonoids, including Oroxylin A Aim: The proposed aimed to investigate in-vitro anti-oxidant activity, toxicity studies and in-vitro cardio protective activity of Oroxylin-A against Doxorubicin mediated myocardial damage on H9c2. Methods: The total phenolic content was estimated using Folin-Ciocalteu test and in-vitro activity was performed using DPPH assay, acute toxicity studies were performed by OECD 423 guidelines. In vitro cardioprotective activity was performed on H9c2 cells and estimated for the biomarkers. Results: Oroxylin-A showed good antioxidant activity. No abnormalities were found in animals upon its usage indicating that Oroxylin-A was safe at 2000 mg/kg.150ug/ml of Oroxylin-A significantly increase cell viability up to 99% and also decreased the LDH and ROS generation which indicates that Oroxylin-A showed significant cardio protective activity on H9c2 cells. Conclusion: This research underscores the potential of Oroxylin A as a candidate for further investigation as cardio-protective agent. Also, the present study contributes to growing body of knowledge aimed at identifying natural compounds that may offer protective effects against myocardial damage, providing hope for future therapeutic interventions in the field of cardiovascular medicine.

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